

Supplementary Material for AGIDefect-4K

This document details the methodologies, resources, and analyses that support our main work, providing a more comprehensive understanding.

This document contains the following core sections:

- **Detailed Defect Definitions:** A clear articulation of the criteria used to classify *Structural Defects* and *Semantic Logic Defects*.
- **Specific Models and Versions:** A comprehensive table listing the closed-source image generation models utilized in our study.
- **Additional Experimental Results:** Quantitative performance of our proposed method implemented with different Multimodal Large Language Model (MLLM) backbones, demonstrating its robustness and generalizability.
- **In-Depth Case Studies:** An analysis of our model’s performance on both flawless and defective images, complete with visual examples and comparative analyses against ground truth annotations.
- **Detailed Annotation Methodology:** An explanation of our rigorous three-stage subjective experiment, covering data classification, annotation, and the double-check validation process.
- **Complete Prompts for Dataset Construction and Evaluation:** The precise prompts used to consolidate human expert annotations and to guide and evaluate the defect description capabilities of MLLMs, along with the detailed scoring rubrics used for the assessment.
- **Licensing Information:** Details regarding the open-access license for the AGIDefect-4K dataset and its components.

The complete **AGIDefect-4K** dataset, including all images, defect masks, and corresponding human-expert annotations, is released on <https://figshare.com/s/6b3ec7e4a90e12a437ac>

1 DEFINITIONS OF DEFECTS

Defects encompass two primary types: Structural Defects and Semantic Logic Defects.

A Structural Defect is defined as an observable error in an image where elements violate the fundamental laws of physical space, anatomy, or geometry. This definition applies under the precondition that the image is not explicitly intended as abstract or surreal art. Specific examples include:

- Anatomical errors (e.g., misplaced limbs, incorrect number of fingers).
- Garbled or nonsensical text.
- Incompletely rendered objects.
- Contradictory perspectives.
- Distorted or inconsistent proportions.

Semantic Logic Defect is defined as a contradiction arising from the combination of elements in an image, violating real-world common sense or domain-specific knowledge. This definition is applicable under the precondition that a surreal or fantastical scene was not explicitly requested. Specific examples include:

- Fire burning underwater.
- Penguins appearing in a desert.
- Out-of-season vegetation (e.g., blooming flowers in a snowy field).
- The presence of a shadow without a corresponding light source.

2 SPECIFIC MODELS AND VERSIONS

The closed-source image generation models used are listed below.

Table 1: Closed-Source Image Generation Models

Vendor	Image Generation Model
OpenAI	gpt-image-1, dalle 3
Midjourney	mijourney_fast
FLUX AI	flux1.1pro
Google	gemini-2.0-flash-preview-image-generation
Ideogram	ideogram_generate_V_1
Kling	kling-v1
xAI	grok-3

3 RESULTS WITH ADDITIONAL BACKBONES

As stated in the main paper, our primary framework AGIDA is implemented based on Qwen3-VL-4B. To further demonstrate the robustness and generalizability of our approach across different architectures, we conducted additional experiments utilizing alternative Multimodal Large Language Model (MLLM) backbones.

Table 2 presents the quantitative results of these experiments. All additional models were fine-tuned under the identical experimental settings described in the main text (e.g., LoRA rank $r = 32$, $\alpha = 64$, AdamW optimizer, and the same 8:2 train-test split). The results indicate that our method maintains consistent effectiveness across various foundational architectures.

Table 2: Performance comparison of AGIDA implemented with different MLLM backbones.

Model	Detection	Explanation		
	AUC	Comp.	Prec.	Mean
Qwen3-VL-4B	0.65	0.89	0.95	0.92
Qwen3-VL-2B	0.68	0.88	0.93	0.90
Qwen2.5-VL-7B	0.69	0.87	0.93	0.90
InternVL3-8B	0.72	0.94	0.91	0.93
InternVL3-2B	0.68	0.88	0.92	0.90

4 CASE STUDIES

To comprehensively evaluate the capabilities and limitations of our proposed model, AGIDA, we conduct in-depth case studies

GOOD CASE

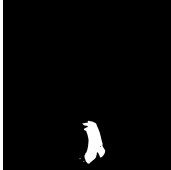
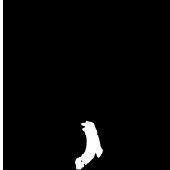
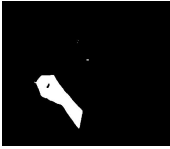
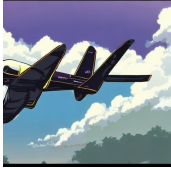
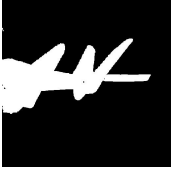
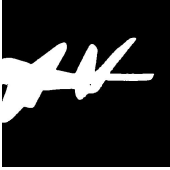
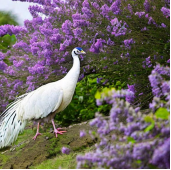
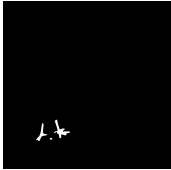
Origin	Description	Origin	Description
	Prompt: A violet rose, with its stem and leaves. AGIDA: The image is flawless. 😊		Prompt: overhead food photography, Bibimbap, Korean restaurant AGIDA: The image is flawless. 😊
	Prompt: A beautiful forest meadow AGIDA: The image is flawless. 😊		Prompt: photograph of a handsome man smiling AGIDA: The image is flawless. 😊
Origin	GTMask	AGIDA	Description
Prompt: symmetry!! ... azure dragon in the east in the fog of magic light, beautiful oriental teen princess... digital painting... 8k 			GT_description: The girl's right hand appears slightly twisted and exhibits structural anomalies... AGIDA: ...a noticeable asymmetry in her hands... This imbalance creates a sense of disharmony in the image... 😊 Claude-Opus-4-6: The image demonstrates excellent generation quality, free from any visual flaws 😊
Prompt: floral textile A letter 			GT_description: The girl's right hand appears slightly twisted and exhibits structural anomalies... AGIDA: The woman's right hand exhibits structural anomalies with subtly fused fingers... 😊 GPT-5.4: ...a near-total prompt mismatch... shows a stylized woman sitting indoors, not a floral textile... 😞
Prompt: bird, driving aircraft, wild, anime style 			GT_description: ...an airplane with significant structural anomalies... The wings... are intertwined in a chaotic manner, and ... AGIDA: ...a fighter jet with structural anomalies. The tail section... is incomplete, and the wings are twisted and deformed... 😊 Qwen3.5-Plus: The image is flawless. 😊
Prompt: a white peacock in a lilac tree 			GT_description: ...feet positioned unnaturally far apart... subtle structural anomaly... slight blur in the foreground... impact on the image's integrity is minimal... AGIDA: ...feet positioned unusually wide and splayed outward... subtle structural flaw in the bird's body... slight blurring... minimally impact the image's integrity... 😊 Kimi-K2.5: This picture is perfect and flawless. 😞

Figure 1: Good Cases: Successful evaluations by AGIDA on both flawless and defective images, demonstrating superior structural defect detection and severity alignment compared to baselines.

on both successful instances (Good Cases) and failure instances (Failed Cases). These cases cover both flawless and defective images, providing a comparative analysis against Ground Truth (GT) annotations and other state-of-the-art Multimodal Large Language Models (MLLMs).

4.1 Successful Cases (Good Cases)

As illustrated in Figure 1, AGIDA demonstrates exceptional proficiency in both verifying high-quality images and identifying complex defects.

Flawless Image Verification: For high-quality generated images, such as the violet rose, overhead food photography, and the forest meadow, AGIDA correctly evaluates them as “flawless,” showing robustness against false positives.

Precise Structural Defect Detection: When analyzing defective images, AGIDA exhibits a high degree of alignment with human expert annotations (GT), significantly outperforming baseline MLLMs.

- *Overcoming Prompt-Bias and Blind Spots:* In the “oriental teen princess” and “floral textile” examples, AGIDA accurately detects subtle structural anomalies (e.g., asymmetrical hands, fused fingers). In contrast, baseline models exhibit severe evaluation blind spots. For instance, Claude-Opus-4-6 completely misses the twisted hand, declaring the image “free from any visual flaws.” Similarly, GPT-5.4 acts merely as a “prompt-checker,” generating a lengthy critique about how the image does not match the “floral textile” prompt, while remaining completely blind to the severely deformed hand in the center of the image. AGIDA successfully avoids this prompt-bias, prioritizing genuine visual artifacts.
- *Alignment on Defect Severity:* In the “white peacock” and “bird driving aircraft” examples, AGIDA perfectly aligns with the GT not only in locating the structural flaws (e.g., twisted wings, unusually wide feet) but also in assessing their severity. It correctly identifies the peacock’s flaws as having a “minimal impact,” demonstrating a human-like tolerance for subtle imperfections, unlike other VLMs that often over-penalize minor defects.

4.2 Failure Cases (Failed Cases)

Despite its strong performance, AGIDA still exhibits certain limitations inherent to current VLM architectures, as shown in Figure 2. Analyzing these failures provides valuable insights for future improvements.

Struggles with Semantic Logic and Common Sense: The most notable limitation occurs when evaluating semantic logic and physical contradictions, where AGIDA lacks top-down reasoning capabilities.

- *Missing Macro-Logic for Micro-Details:* In the “sled dog” example, the GT correctly identifies a massive logical error: the dog is facing the sled instead of pulling it, contradicting fundamental real-world physics. AGIDA completely misses this macro-level semantic logic defect. Instead, it employs a “bottom-up” pixel-level scanning strategy, over-fixating on micro-details and hallucinating anatomical errors (e.g.,

“left eye disproportionately small,” “right hind leg positioned incorrectly”).

- *Insensitivity to Subtle Physical Violations:* Similarly, in the “broken light bulb” case, a flower emerges unnaturally from the side of the glass. AGIDA fails to recognize this subtle violation of real-world physical logic and incorrectly classifies the image as “flawless.” Both cases demonstrate that while the model excels at finding structural deformations, it struggles to comprehend the logical relationship between objects.

Omissions and Hallucinations in Fine-Grained Details:

AGIDA also shows vulnerabilities when defects are highly subtle, integrated into the artistic style, or located in ambiguous regions.

- *Missing Stylized Artifacts:* In the cyberpunk street scene, the GT identifies garbled text. However, because these text artifacts heavily resemble neon lights and integrate well with the overall artistic style, AGIDA’s fine-grained text recognition (OCR) fails, leading it to miss the defect and declare the image “flawless.”
- *Ignoring Background Degradation:* In the “great old ones sitting in a bar” example, the GT highlights severe background degradation (glasses and items blurred into an indistinguishable mass). AGIDA completely misses these subtle background flaws. Instead, it hallucinates structural anomalies in the main subject, claiming the dog has an “incomplete body.”

These failure cases highlight a critical bottleneck in current visual evaluation models: while they are becoming highly adept at detecting pixel-level structural distortions in main subjects, they still lack the top-down common-sense reasoning required to identify macro-level semantic contradictions, as well as the fine-grained perception needed to distinguish genuine stylistic choices from subtle text or background artifacts.

5 SUBJECTIVE EXPERIMENT SETTINGS

Our subjective experiment incorporates a rigorous three-stage process: **data classification**, **data annotation**, and **double-check validation**.

In order to classify each image, consensus among participants is required. If participants unanimously agree that an image has defects, it is classified as ‘Reject’, indicating that the image has defects. Conversely, if the image is agreed to be without defects, it is classified as ‘Accept’. In cases where there is no unanimous agreement, expert intervention is needed to provide a definitive judgment. The interface screenshot for classification is illustrated in Figure 3, where the ‘Accept’ button represents an image deemed to have no defects, and the ‘Reject’ button signifies an image judged to have defects.

In the subsequent data annotation stage (detailed in Fig. 4), participants are required to meticulously examine each image alongside its associated generation prompt. Their primary tasks include identifying any defects, generating precise mask annotations for these areas using the Segment Anything Model (SAM), and providing detailed descriptions that cover the defect’s location within the image, its classification, and the potential impact on overall image quality.

FAIED CASE

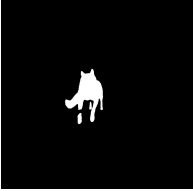
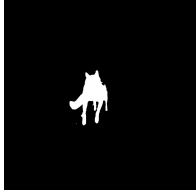
Origin	Description	Origin	Description
	GT_description: ...text that appears to be garbled, resembling both lights and letters... slightly reduces the overall quality, yet it integrates well with the style... AGIDA: The image is flawless. 😞		GT_description: ...flower emerging from the left side of a broken light bulb... extra part on top... effect on the image's quality is minimal. AGIDA: The image is flawless. 😞
Origin	GTMask	AGIDA	Description
Prompt: ...a sled dog is pulling a red sled, gliding on a white snowy ground... with powerful color contrast...			
			GT_description: ...clear directional error with the sled dog... image shows the sled dog facing directly towards the sled, which... contradicts the common logic... AGIDA: ...exhibits several notable defects. The dog's eyes appear unnatural, with the left eye being disproportionately small and the right eye lacking a pupil... Additionally, the dog's right hind leg is positioned incorrectly... which is anatomically incorrect... 😞
Prompt: cute great old ones sitting in a bar, warm color			
			GT_description: ...dog on the right side... lacking visible eyes... glasses on the table and the items on the shelf... blurred into an indistinguishable mass... severely compromising the image's clarity... AGIDA: ...dog with significant structural anomalies, including a distorted face and an incomplete body... severely impact the overall quality... 😞

Figure 2: Failed Cases: Instances where AGIDA struggles, including missing text rendering artifacts, failing to recognize subtle background degradation, and lacking the macro-level common sense required to identify severe semantic logic errors (e.g., the reversed sled dog).

In the double-check validation phase, we implement a meticulous review system utilizing three independent reviewers who critically assess all participant annotations, including masks and descriptions. Our interface for double check, shown in Figure 5, allows the reviewers to collaborate effectively. Through this interface, they determine which masks are accurate and can be consolidated into a unified representation. Similarly, they discern the correct descriptions to retain. During this stage, additional checks are conducted for cases where there are discrepancies in the masks and descriptions provided by participants. This rigorous process ensures that only precise and consistent annotations are incorporated into the final dataset, thereby enhancing the overall reliability and quality of the data.

6 PROMPTS FOR DATASET CONSTRUCTION

After accurately curating defect descriptions, we utilize GPT-4o to consolidate multiple descriptions related to the same image. The prompt given to GPT-4o for this task is detailed below:

#System Prompt: You are an expert in assessing the quality of AI-generated images, specializing in identifying defects. You will receive

Chinese descriptions of defects within an image, authored by one or more human experts, alongside the corresponding AI-generated image. Your tasks are:

- **Integrate Descriptions:** Merge the human experts' descriptions of defects within the image, using the image content for reference.
- **Refine Output:** Further refine the consolidated descriptions to produce a polished English output.

Key Considerations:

- The final description must be accurate, professional, and faithfully represent the original human expert descriptions, preserving their intended meaning.
- If multiple descriptions are provided (e.g., from two experts), synthesize common points while retaining unique and valuable defect information. Ensure conciseness by avoiding redundancy but comprehensively cover all critical defects.
- If indicated in the original human expert descriptions, or if a clear inference can be drawn from the defect's severity, briefly

articulate the impact of these defects on the image's overall quality, realism, aesthetic appeal, or thematic expression.

Please provide your response strictly in the following JSON format:

```
{
  "description": "Your defect description..."
}
```

Return only this JSON object, without any additional explanations, analyses, or extraneous text.

#User Prompt: Below is an AI-generated image [image] and its defect description(s) [description block] provided by human expert(s). Based on your observation of the image and the provided human expert defect description(s), please provide a description.

Prompt for Defect Category Classification

To further analyze the distribution of defect types and their relation to perceived quality, we utilize GPT-4 to classify the integrated defect descriptions into five distinct categories. The prompt used for this classification task is detailed below:

#System Prompt: You are an expert data analyst specializing in image defect categorization. Your task is to classify a given image defect description into exactly ONE of the following five categories:

- **body**: Anatomical errors related to the human body (e.g., distorted limbs, extra fingers, awkward postures), excluding the face.
- **faces**: Anomalies specifically located on human faces (e.g., asymmetrical eyes, garbled teeth, distorted facial structures).
- **animals**: Structural or anatomical defects occurring in animals.
- **objects**: Incomplete rendering, structural anomalies, or distortions in inanimate objects, texts, or background elements.
- **logical_defects**: Semantic contradictions, violations of physical laws, or common-sense errors (e.g., floating items, incorrect shadows, fire underwater).

Rule: If a description mentions multiple issues, classify it based on the primary or most severely described defect.

Return ONLY a valid JSON object in the exact format: {"category": "category_name"}. Do not include markdown code blocks or any other text.

#User Prompt: Please classify the following defect description: [defect description]

7 PROMPTS FOR EXPERIMENTS

7.1 Prompts for MLLM Defect Description Generation

To assess the defect description capabilities of multimodal large language models, we developed a framework centered around a comprehensive evaluation process, aimed at guiding the model in crafting accurate defect assessments. Our objective was to prompt the MLLM to generate detailed descriptions of defects in AI-generated images while avoiding over-sensitivity to artistic intent. We provided the following prompt to instruct the model:

#System Prompt: You are an expert in evaluating the quality of AI-generated images. Your task is to determine whether an image contains significant defects.

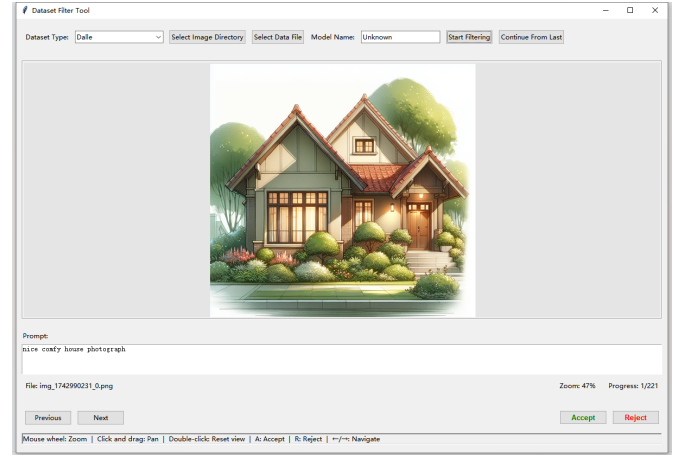


Figure 3: The interface screenshot for classification

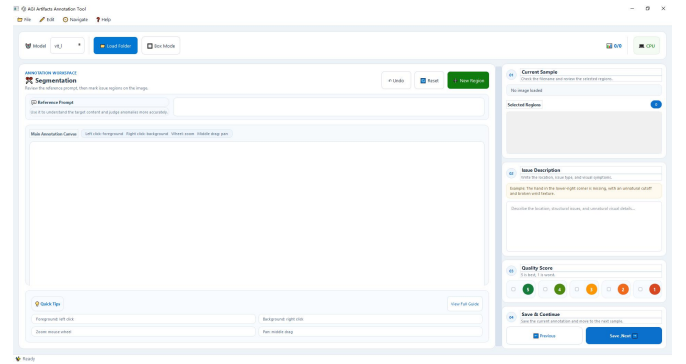


Figure 4: The interface screenshot for annotation

Step 1: Careful Observation

Examine the image holistically. Consider whether the overall composition, anatomy, physics, and logic appear reasonable and coherent.

Step 2: Defect Identification (if applicable)

If you notice potential anomalies, evaluate whether they constitute genuine defects:

- **Structural Defects**: Clearly malformed body parts, impossible anatomy (e.g., six fingers, missing limbs), severely distorted objects.
- **Logical Defects**: Obvious violations of physical laws or common sense that are clearly unintentional.

Do not flag as defects:

- Minor stylistic variations or artistic exaggeration.
- Subtle proportion differences that don't significantly impact visual quality.
- Features that could be intentional artistic choices.

Step 3: Prompt Verification

If defects were identified, check whether the prompt explicitly describes or justifies them. Intentional features described in the prompt are not defects.

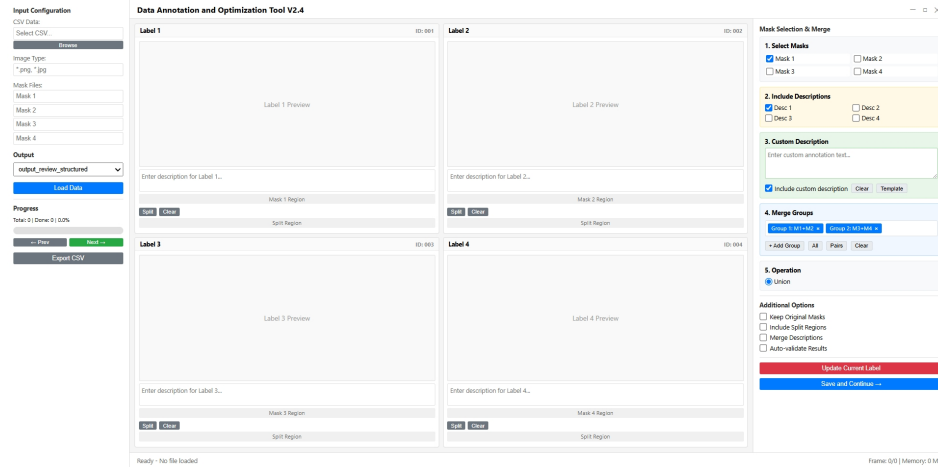


Figure 5: The interface screenshot for double check

Step 4: Final Judgment

Provide your assessment. Be conservative — when uncertain, lean toward classifying the image as not defective.

Provide your assessment in plain JSON format:

```
{
  "Defective": [true/false],
  "Description": []
}
```

If no defects are found, the "Description" field should remain an empty list []. If defects are present, provide a single string entry within the "Description" list that describes all identified defects in detail, covering their locations and visual appearance, and concluding with how they collectively impact the overall quality of the image.

Example 1 (No defect):

Prompt: a golden retriever sitting in a sunny park

```
{
  "Defective": false,
  "Description": []
}
```

Example 2 (With defects):

Prompt: two mouse wearing chefs' hats

```
{
  "Defective": true,
  "Description": [
    "The image depicts two mice with several significant anatomical abnormalities. Both mice have severely malformed toes, appearing excessively numerous and disorganized. The mouse on the left exhibits an unusual mouth shape, while the mouse on the right is missing its right ear. Additionally, there is an extra pair of feet floating unsupported between the two mice. These combined anomalies greatly diminish the overall quality, coherence, and aesthetic appeal of the image, making it appear quite odd."
  ]
}
```

#User Prompt: For each image [image] analyzed, the MLLM (configured with the system prompt above) then receives the following input:

Original generation prompt [generation prompt]

Please assess this image [image] for defects based on the provided prompt.

7.2 Prompts for Evaluating LLMs

In this part of the evaluation, we focus on assessing the MLLMs capability to identify and describe defects. Utilizing a GPT-assisted evaluation strategy, we systematically compare LLM-generated descriptions against human-annotated benchmarks, concentrating on two critical criteria: **Completeness** and **Precision**.

The evaluation's goal is to confirm that LLM-generated defect descriptions are both complete and accurate, aligning with expert human interpretations. Below is a structured approach to this evaluation:

#System Prompt: You are an expert in evaluating AI-generated images for detecting visual defects. Your expertise lies in pinpointing two main types of defects:

- (1) **Structural Defects:** Violations of basic physical space rules (e.g., misaligned limbs, missing/extra fingers, incomplete key objects).
- (2) **Semantic Logic Defects:** Combinations that defy common sense or domain knowledge and are not intended by the prompt (e.g., floating objects without support, physically impossible arrangements).

Prompts for Defect Description Completeness: Evaluate if the description [MLLM DESC] encompasses all visual defects mentioned in the reference description [GOLDEN DESC]. Assess solely on whether each defect in the reference is at least acknowledged in the output, without evaluating the description's accuracy, detail, or relevance.

Rating System for Completeness:

- **Score 2:** All or nearly all defects present in the reference are mentioned.

- **Score 1:** Some, but not all, defects from the reference are mentioned.
- **Score 0:** None of the defects from the reference are mentioned.

Please only provide the result in the following JSON format: {"Score":[]}.
Reference Description [**GOLDEN DESC**] MLLM Description [**MLLM DESC**].

Prompts for Defect Description Precision: The precision metric measures descriptions in terms of defect details (location, type, severity) and penalizes descriptions with controversial or contradictory information compared to the reference.

Rating System for Precision:

- **Score 2:** Minor inconsistencies or less controversial details compared to the reference; generally consistent.
- **Score 1:** Major inconsistencies or more controversial details than the reference; noticeable contradictions.
- **Score 0:** Severe contradictions or completely inaccurate details compared to the reference; highly controversial.

Please only provide the result in the following JSON format: {"Score":[]}.
Reference Description [**GOLDEN DESC**] MLLM Description [**MLLM DESC**].

8 LICENSE

We are committed to facilitating maximum research reproducibility and community engagement by fully and publicly releasing all core components of the **AGIDefect-4K** dataset. This includes all images exhibiting defective masks, the prompts used for their generation (where applicable), and the corresponding detailed human-annotated defect descriptions.

All human-rated data (our defect annotations) will be made available in its entirety, presented in a carefully organized and anonymized format. This commitment to open access is intended to maximize transparency and foster widespread academic progress in the fields of AI-generated image analysis, defect detection, and multimodal foundation model evaluation. We hope our endeavour will significantly support and accelerate academic advancements and the development of innovative solutions by providing the community with this comprehensive resource.